Hidden Markov Modeling of Sleep Stages from Single-Channel EEG Using Spectral Features

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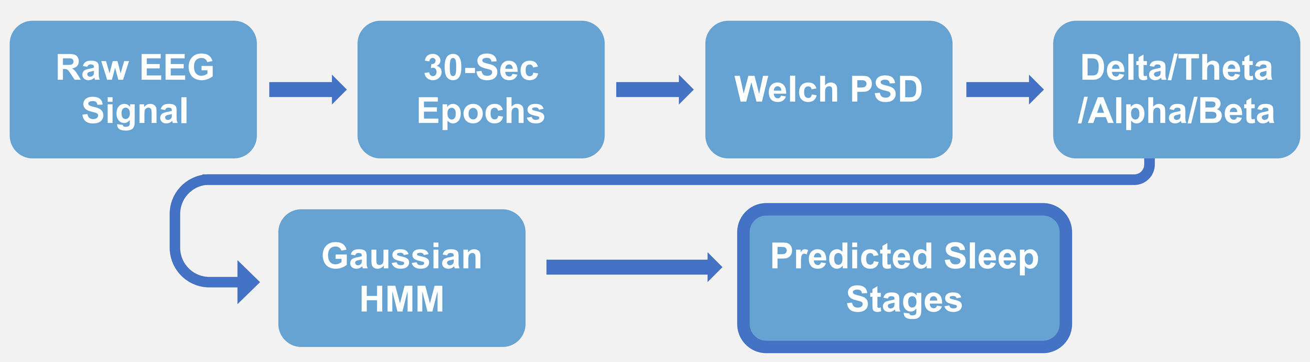
go/sleep-hmm-results/

Overview

Sleep staging is typically performed by manually assigning 30-second EEG epochs to Wake, NREM, or REM sleep stages.[#] Since EEG frequency structure changes across sleep stages and transitions occur non-randomly over time, Hidden Markov Models (HMMs) are well suited for modeling sleep architecture.



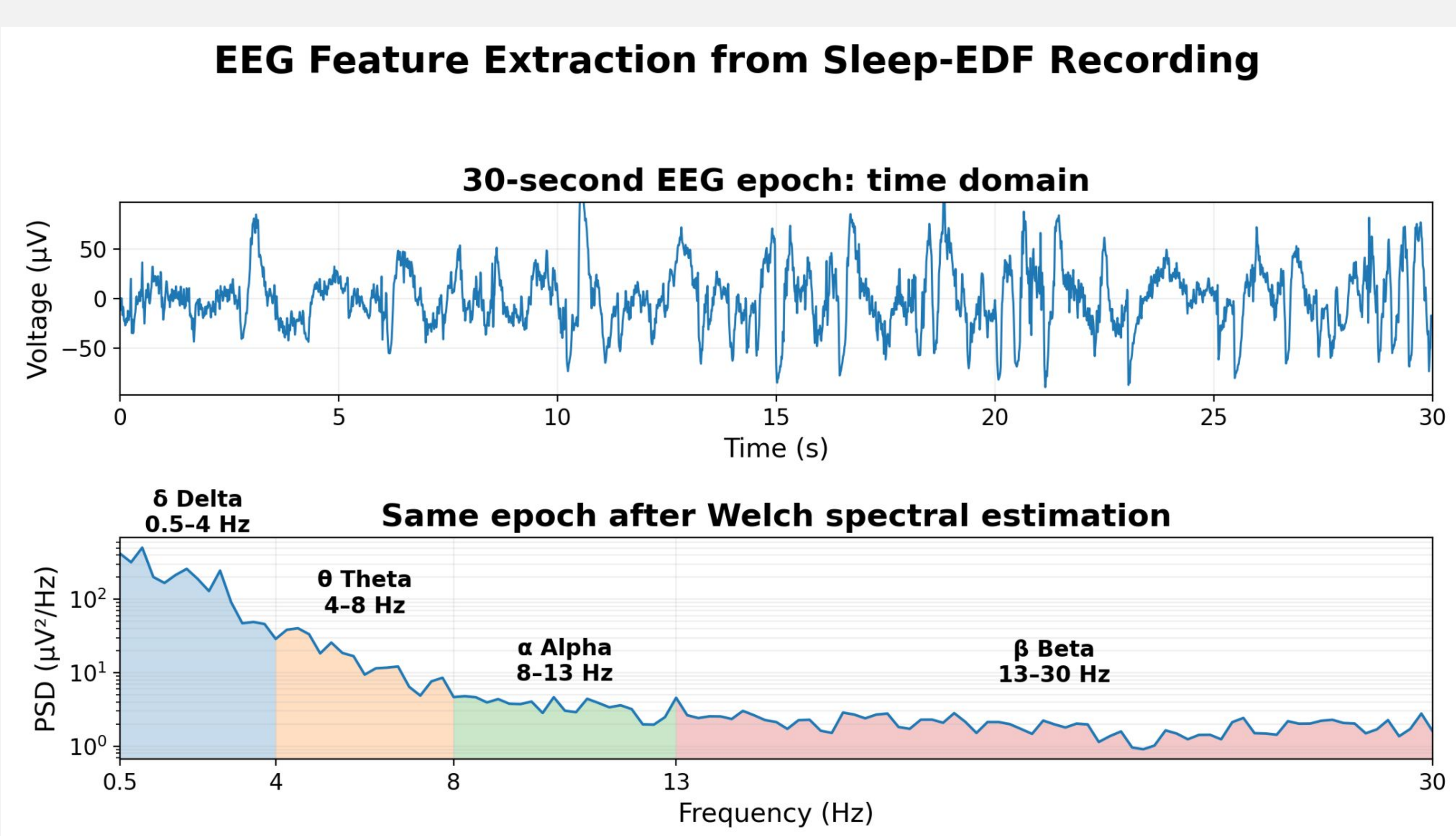
This project evaluated whether a Gaussian HMM could recover sleep stages from single-channel EEG* recordings in the Sleep-EDF Expanded dataset using spectral band-power features derived from Welch power spectral density estimation.

Background + Dataset

Wakefulness is associated with increased alpha and beta activity, while NREM sleep is dominated by delta oscillations. REM[$] sleep exhibits lower-amplitude mixed-frequency activity with elevated theta power relative to NREM.

Single-channel Fpz–Cz[@] EEG recordings and expert sleep-stage annotations were obtained from the publicly available Sleep-EDF Expanded dataset hosted on PhysioNet[2]. Recordings were segmented into consecutive 30-second epochs following standard sleep-scoring conventions.[1]

To simplify the classification problem, N1, N2, N3, and N4 labels were merged into a single NREM state.



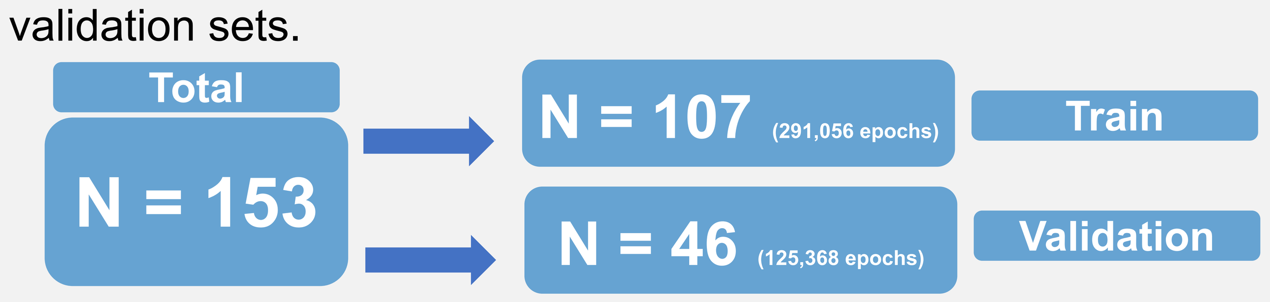
Feature Extraction + HMM

EEG recordings were divided into 30-second epochs and transformed into delta, theta, alpha, and beta relative band-power features using SciPy's implementation of Welch spectral estimation. EEG band-power ranges were based on prior sleep literature.[1]

These features were used as observations for a three-state, fully connected Gaussian HMM representing: Wake, NREM, and REM.

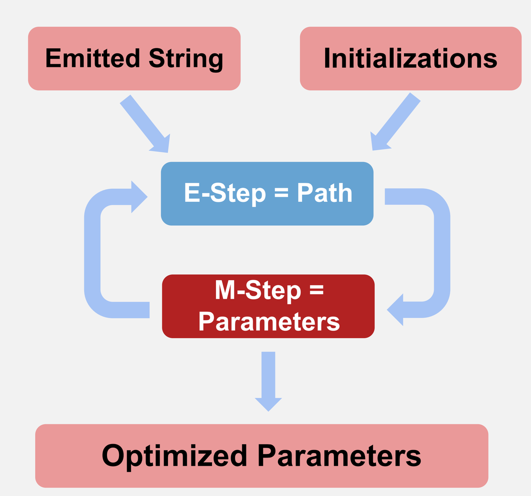
Data Splitting

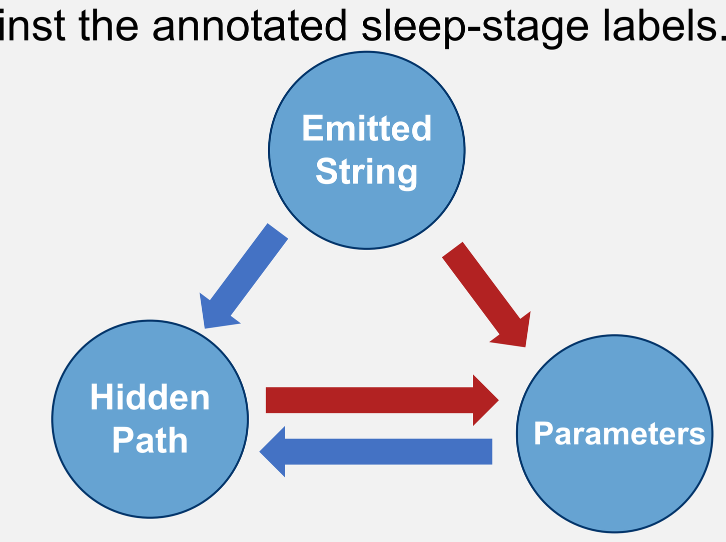
Participants were randomly split into training and validation groups: 70% for training and 30% for validation. Recordings from same participant (but different night) were kept together in the same group. Every experiment used the same training and validation sets.



Training + Decoding

Model parameters were learned using the Baum–Welch expectation-maximization algorithm. We compared uniform, random, and biologically informed transition initialization strategies, then used Viterbi decoding during validation to assign the most likely sleep-stage sequence and compare predicted states against the annotated sleep-stage labels.

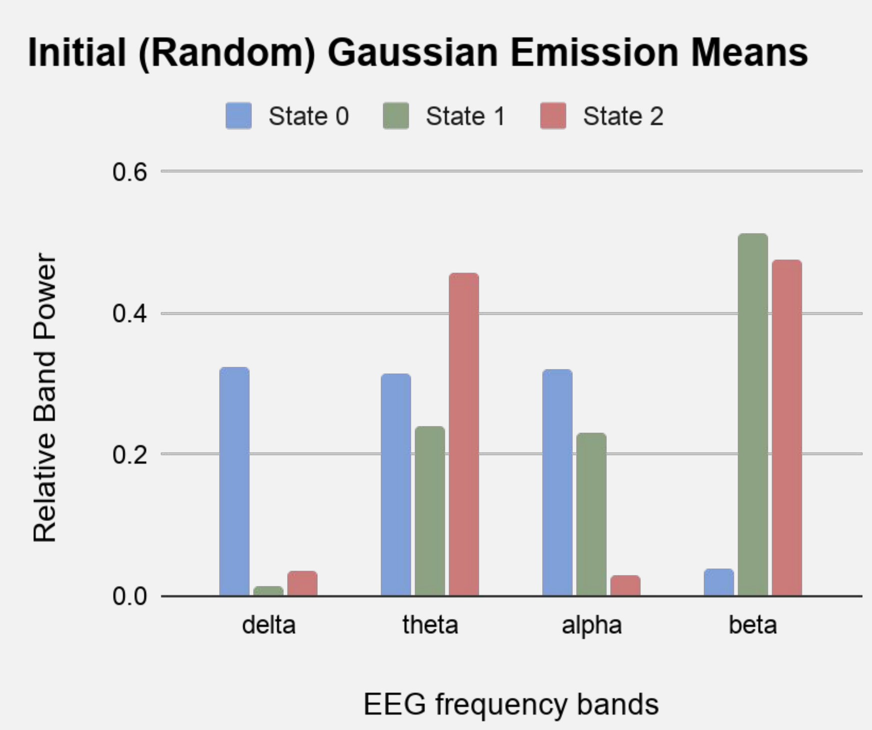




For the highest-performing run, (see *Results*) a biologically informed transition matrix was initialized based on prior literature and with high self-transition probabilities because sleep stages are temporally persistent across consecutive 30-second epochs.[4] Gaussian emission means were initialized using random values. Gaussian variances were uniform.

Initial Transition Matrix (Biologically Informed)

	WAKE	NREM	REM
WAKE	0.945	0.050	0.005
NREM	0.015	0.980	0.005
REM	0.005	0.030	0.965



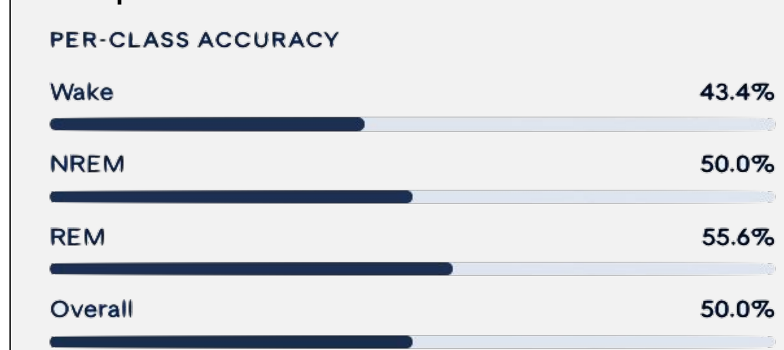
We additionally compared relative-power and log-relative-power feature representations to evaluate how feature representation affected model performance. Relative-power features used smaller uniform initial Gaussian variances (.05) because values are bounded between 0 and 1, while log-relative features required larger initial variances due to their wider numerical range (.75).

Results

Eight model configurations were evaluated across combinations of:

- **Features:** relative vs. log-relative spectral
- **Transition Initialization:** informed vs. uniform vs. random
- **Emission Initialization:** informed vs. random

Run 07 — relative features, informed transition initialization, random emission initialization, and uniform Gaussian variances — produced the strongest overall performance with 50.0% classification accuracy. Most other configurations collapsed into two effective states or failed to separate Wake and REM sleep.

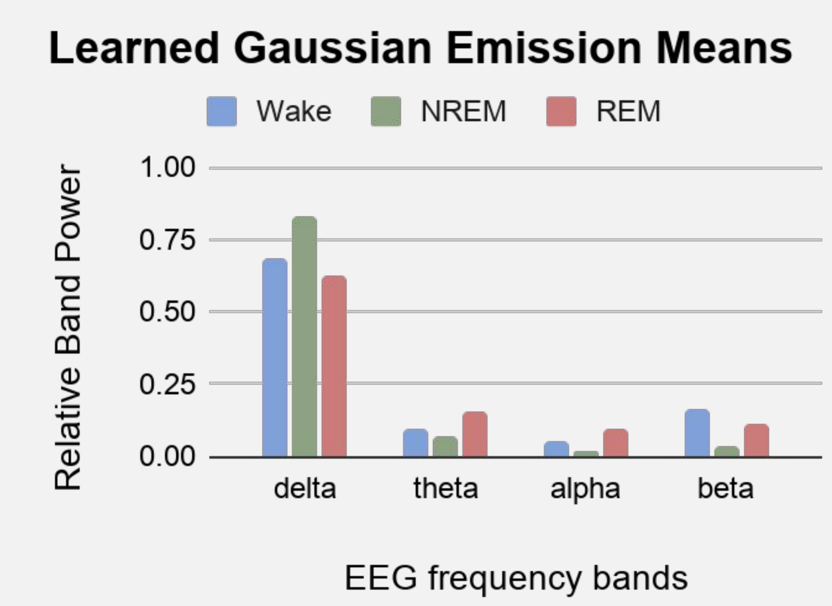


REM achieved the highest per-class accuracy (55.6%), followed by NREM (50.0%). Wake was the worst-performing class at 43.4%, with the majority of its epochs misclassified as NREM.

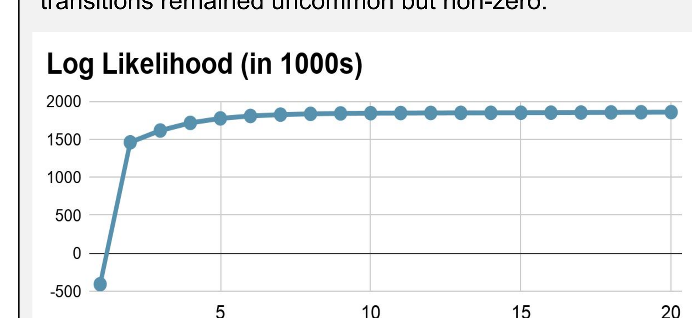
Learned Transition Matrix

	WAKE	NREM	REM
WAKE	0.861 (-0.084)	0.125 (+0.075)	0.013 (+0.008)
NREM	0.066 (+0.051)	0.917 (-0.063)	0.017 (+0.012)
REM	0.027 (+0.022)	0.059 (+0.029)	0.913 (-0.052)

All three states developed strong self-transition probabilities (0.861–0.917), consistent with the temporal persistence of sleep stages.The most common cross-stage transitions were: Wake → NREM, NREM → Wake. Direct Wake → REM transitions remained uncommon but non-zero.



NREM exhibited the strongest delta power (0.828). Wake showed elevated beta activity (0.162). REM displayed the highest theta (0.155) and alpha (0.096) power of the three states alongside reduced delta power.



Log-likelihood increased across all 20 iterations without plateauing, indicating the algorithm had not fully converged.

Confusion Matrix

True \ Pred	Wake	NREM	REM
Wake	37,206	43,043	5,497
NREM	10,309	15,747	5,422
REM	1,503	1,953	4,329

Concurrent with the above Per-Class Accuracy, REM had the highest accuracy across the states, followed by NREM. The highest levels of mis-identification were found between Wake and NREM, with 50.2% of true Wake states being predicted as NREM, and 32% of true NREM being predicted as Wake.

Discussion

Initialization drives model success

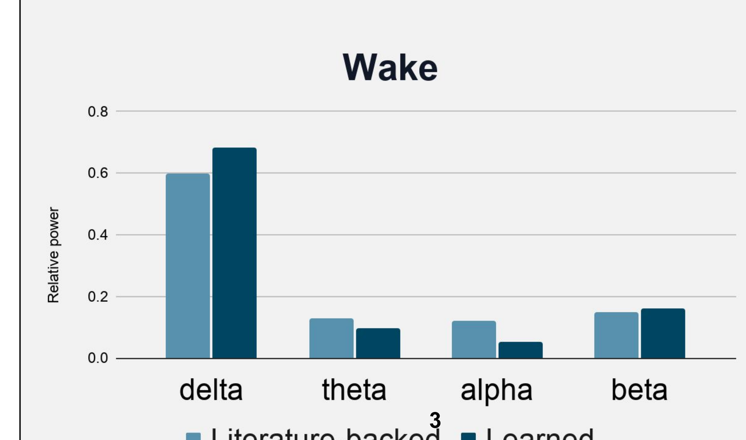
Across the log-relative experiments, the Gaussian variance collapsed close to 0, resulting in models that never predicted REM. Relative features combined with biologically informed transition initialization and random emission initialization was the only configuration that predicted all three sleep stages.

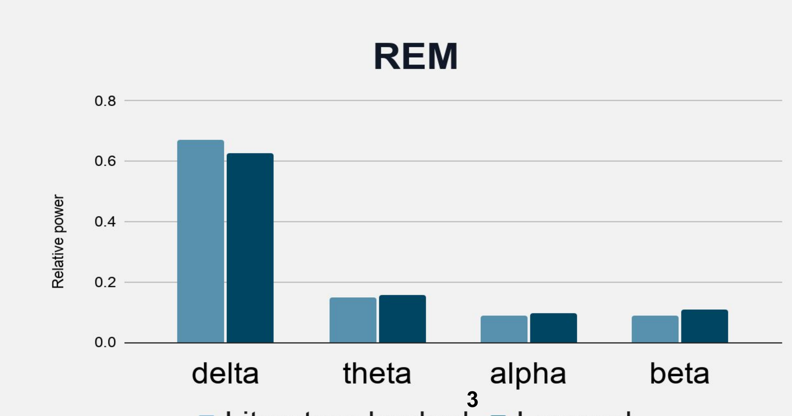
Wake and NREM model confusion

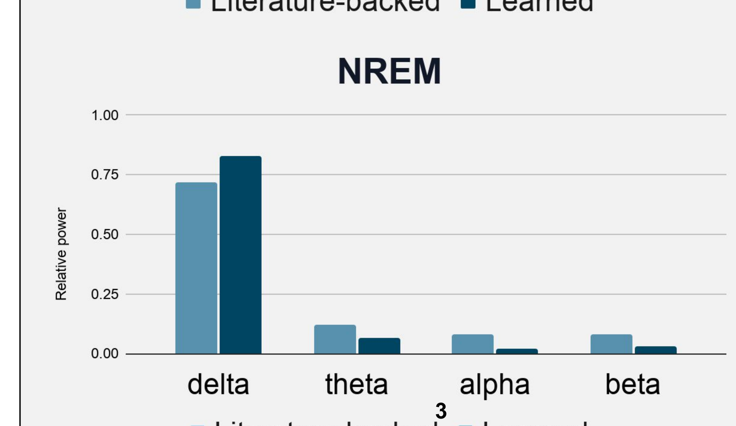
Wake was often misclassified as NREM likely because the learned transition matrix favored Wake→NREM much more than Wake→REM. In run 07, Wake→NREM was 0.125, while Wake→REM was only 0.013.

Learned gaussian means align with literature

The learned Gaussian means aligned with known EEG physiology: NREM showed dominant delta power (0.83), Wake showed elevated beta (0.16), and REM exhibited the highest theta and alpha of the three states — without any labeled training data.[3]







ELSI: This project used publicly available, de-identified Sleep-EDF data and collected no new human data. Automated sleep staging could reduce manual scoring time and improve access to sleep analysis, but inaccurate predictions could affect diagnosis. Because EEG patterns vary across individuals and recording conditions, models like this should support, not replace, expert sleep-stage scoring.

References

1. Iber, C., & American Academy of Sleep Medicine. (2007). The AASM manual for the scoring of sleep and associated events: Rules, terminology and technical specifications. American Academy of Sleep Medicine.
2. Goldberger, A., Amaral, L., Glass, L., Hausdorff, J., Ivanov, P. C., Mark, R., & Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation, 101(23), e215–e220. https://doi.org/10.1161/01.CIR.101.23.e215
3. Hussain, I., Hossain, M. A., Jany, R., Bari, M. A., Uddin, M., Kamal, A. R. M., Ku, Y., & Kim, J.-S. (2022). Quantitative evaluation of EEG-biomarkers for prediction of sleep stages. Sensors, 22(8), 3079. https://doi.org/10.3390/s22083079
4. Metzner, C., Schilling, A., Traxdorf, M., Krauss, P., & Schulze, H. (2021). Sleep as a random walk: A super-statistical analysis of EEG data across sleep stages. Communications Biology, 4, 1385. https://doi.org/10.1038/s42003-021-02912-6
5. Kemp, B., Zwinderman, A. H., Tuk, B., Kamphuisen, H. A. C., & Oberye, J. J. L. (2000). Analysis of a sleep-dependent neuronal feedback loop: The slow-wave microcontinuity of the EEG. IEEE Transactions on Biomedical Engineering, 47(9), 1185–1194. https://doi.org/10.1109/10.867928
6. Silber, M. H., Ancoli-Israel, S., Bonnet, M. H., Chokroverty, S., Grigg-Damberger, M. M., Hirshkowitz, M., Kapen, S., Keenan, S. A., Kryger, M. H., Penzel, T., Pressman, M. R., & Iber, C. (2007). The visual scoring of sleep in adults. Journal of Clinical Sleep Medicine, 3(2), 121–131. https://doi.org/10.5664/jcsm.26814

- standardized 30-second interval; * - electroencephalogram; $ - REM: Rapid Eye Movement; @- frontal-to-central EEG channel;